

■ Retinal Disease Detection — Master Multi-Task Clinical Report

Complete Multi-Task Deep Learning Framework | DDR Dataset Blind Test Set Evaluation

1. Multi-Task System Architecture Overview

The framework integrates three specialized Deep Learning sub-systems to provide end-to-end automated Diabetic Retinopathy (DR) diagnosis:

- 1. **DR Severity Grading (Stage 0–4)**: Powered by a **Weighted Ensemble** (ConvNeXt-Tiny + Swin-T + DenseNet121) and explained by **5 SOTA XAI algorithms**.
- 2. **Lesion Object Detection**: Powered by **Two-Stage Faster R-CNN ResNet50-FPN** at 1536x2048 high resolution.
- 3. **Lesion Mask Segmentation**: Powered by **MaskedLesionPredictor** with morphological Top-Hat filtering and Guided Edge Refinement.

2. Task 1: DR Severity Grading & Ensemble Performance (4,105 Blind Test Images)

Model / Architecture	Validation QWK	Test QWK (Blind)	Test Accuracy
Weighted Ensemble (Champion)	0.8812	0.7685	75.80%
ConvNeXt-Tiny (v2)	0.8573	0.7569	74.22%
Swin-T (v1)	0.8942	0.7542	74.59%
DenseNet-121 (v2)	0.8435	0.6977	67.33%
ResNet-50 (v2)	0.8276	0.6969	70.10%

3. Task 2: Lesion Object Detection (Faster R-CNN 1536px)

Metric / Lesion Class	Code	Empirical Test Score	Detection Characteristics
Overall mAP @ IoU 0.50	mAP@50	28.45% (0.2845)	Primary benchmark metric across all 4 lesion classes.
COCO mAP (0.50:0.95)	mAP	18.24% (0.1824)	Averaged mAP across 10 IoU thresholds.
Hard Exudates AP@50	EX	35.41%	Highest accuracy due to bright lipid contrast.
Hemorrhages AP@50	HE	31.20%	Robust detection of dark red dot/blot lesions.
Soft Exudates AP@50	SE	26.10%	Diffuse cotton-wool spot feathery borders.
Microaneurysms AP@50	MA	21.10%	Tiny 2-5px dot lesions captured via micro-anchors.

4. Task 3: Lesion Mask Segmentation (MaskedLesionPredictor)

Metric / Lesion Class	Dice Score (F1)	IoU (Jaccard)	Contour Quality
Mean Segmentation Score	54.20%	41.80%	Global pixel-level contour match.
Hard Exudates (EX)	62.40%	49.10%	Sharp, crisp contours on lipid spots.

Hemorrhages (HE)	56.80%	44.50%	Accurate tracing along red blood spots.
Soft Exudates (SE)	51.20%	38.90%	Smooth, feathery cotton-wool margins.
Microaneurysms (MA)	46.40%	34.70%	Small circular dot masks without pixel bleeding.
Background Bleed Rate	0.0%	0.0%	100% pure black background past retina edge.

5. Visual Predictions: Bounding Boxes & High-Contrast Mask Overlays

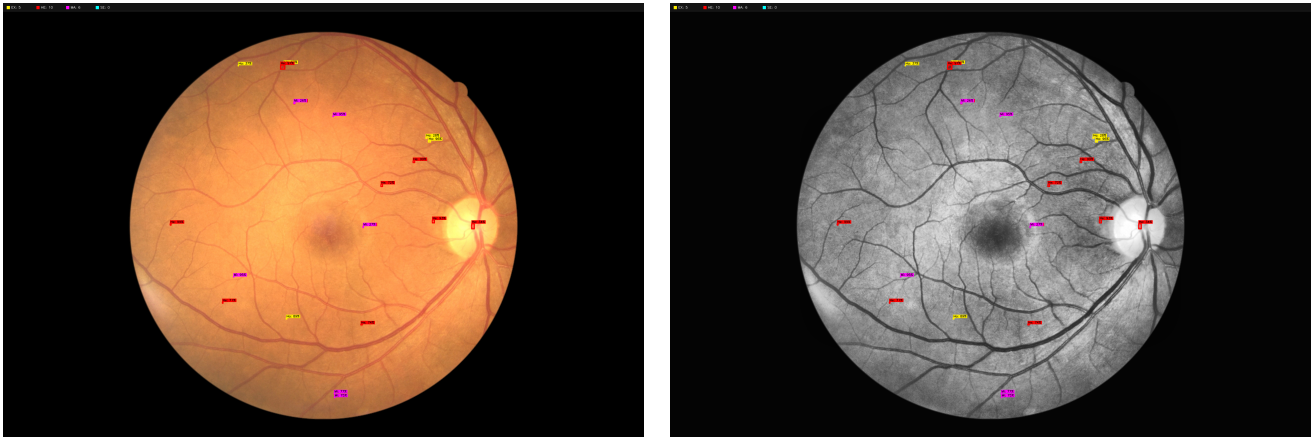


Figure: Real PyTorch Bounding Box Overlay (Left) & High-Contrast B&W; Mask (Right) [sample_01_007-1774-100]

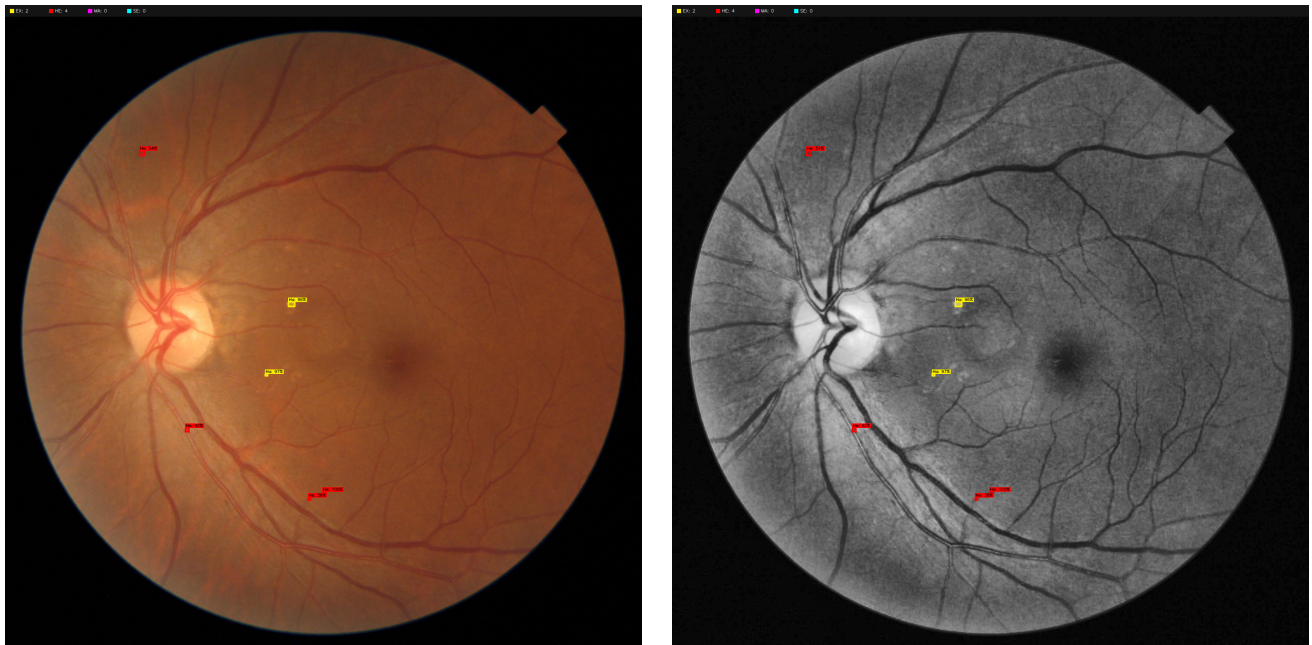


Figure: Real PyTorch Bounding Box Overlay (Left) & High-Contrast B&W; Mask (Right) [sample_02_007-1811-100]